

Product Line Profitability & Margin Performance Analysis for Nassau Candy Distributor

Research Paper / Executive Analytical Report

Executive Summary

Across 10,194 order-line records, total sales are \$141,783.63, total gross profit is \$93,442.80, and weighted gross margin is 65.91%. Chocolate is the dominant profit engine, contributing 95.06% of gross profit. Five products account for approximately 95.06% of total gross profit, creating significant portfolio concentration.

1. Data Cleaning & Validation

The dataset contains 10,194 rows and 18 fields with no missing values. Sales, units and cost are positive, and gross profit reconciles exactly to Sales minus Cost. Order dates span 2024–2025. Ship dates span 2026–2030, creating implausibly long shipment intervals; shipment timing is therefore excluded from profitability conclusions and should be validated.

2. KPI Framework

Gross margin is $\text{Gross Profit} / \text{Sales}$. Profit per unit is $\text{Gross Profit} / \text{Units}$. Revenue and profit contribution measure each product's share of total sales and total profit. Overall weighted gross margin is 65.91%.

3. Product-Level Profitability

The five Chocolate Wonka Bar products dominate both sales and profit. Scrumdiddlyumptious has the highest gross profit (\$19,357.50), followed by Triple Dazzle Caramel (\$18,610.20), Milk Chocolate (\$17,443.37), Nutty Crunch Surprise (\$16,819.95), and Fudge Mallows (\$16,593.60). Their margins range from 64.92% to 71.35%.

4. Division Performance

Chocolate produces 92.88% of revenue and 95.06% of profit at 67.45% margin. Other produces 6.82% of revenue but only 4.64% of profit at 44.84% margin. Sugar contributes 0.30% of both revenue and profit with a 66.61% margin. The Other division is the clearest structural margin concern.

5. Pareto / Profit Concentration

The top four products generate 77.30% of profit, while the top five generate 95.06%. This demonstrates high dependency on a small group of Chocolate products and suggests that availability, pricing and sourcing for these products should be actively protected.

6. Margin Risk

Kazookles is the most significant margin-risk product with \$1,205.75 sales, \$92.75 gross profit and 7.69% margin. Other sub-50% products include Fun Dip (40%), Nerds (46.67%), SweeTARTS (46.67%), and Lickable Wallpaper (50%). These products warrant pricing, sourcing or assortment review.

7. Margin Volatility

Monthly gross margin is comparatively stable. The standard deviation of monthly gross margin is approximately 0.53 percentage points, suggesting product mix and division structure are more important than large month-to-month margin swings.

8. Recommendations

Prioritize the five core Chocolate products; review Kazookles and sub-50% margin products; investigate sourcing and pricing in Other; use profit contribution alongside sales volume; and validate shipment dates before any route or logistics analysis.

9. Limitations

The dataset has only 15 products, so profit concentration is inherently high. No promotion, supplier, inventory, target-margin or price-change fields are supplied, so the analysis identifies priorities rather than proving causal drivers.

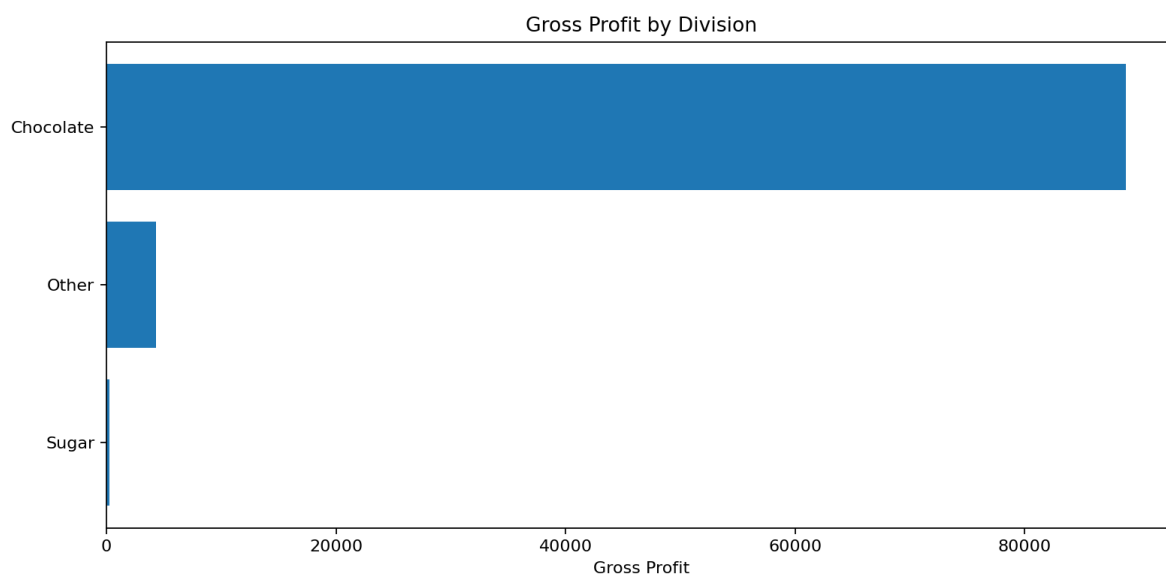
10. Conclusion

The business is strongly profitable overall, but profitability is concentrated in Chocolate while the Other division and Kazookles present clear margin risks. A filterable dashboard can support product rationalization, pricing review and portfolio monitoring.

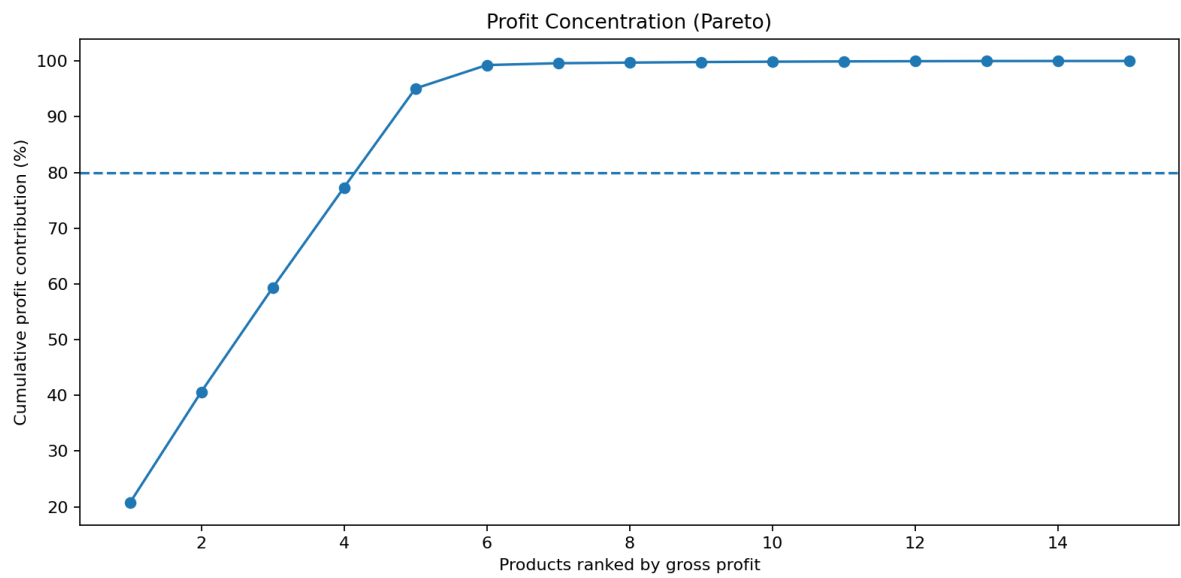
Division Metrics

Division	Sales	Profit	Margin	Profit %
Chocolate	\$131,692.90	\$88,824.62	67.45%	95.06%
Other	\$9,663.25	\$4,333.45	44.84%	4.64%
Sugar	\$427.48	\$284.73	66.61%	0.30%

Gross Profit by Division



Profit Concentration



Cost vs Sales Diagnostics

